# THREE RELATIONS HOLDING IN THE DEMAZURE MODULE ATTACHED TO A MULTIPLE OF THE HIGHEST ROOT

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*Allowed standard prerequisites (editorial).*

- The local Weyl module W(lambda) and its quotient the Demazure module D(1, lambda), with the defining relations (w2) and (w2') remembered as the annihilation of the elements x^-_alpha t^m acting on the highest weight vector. The author's sections 2 and 3, in which the modules and the relations are set up before the counted statement. Fixes the module D(1, (k+1)theta) in which the three relations of the statement are asserted.
- The copy of sl_2 spanned by x^+_alpha, x^-_alpha and alpha^dual inside the current algebra, and the standard sl_2 weight argument. Used inside the counted proof at source lines 469-476. Produces part (1) from the vanishing of the two vectors displayed in the proof.
- The commutation relation for two negative root vectors acting on the highest weight vector, labelled garland in the article. Invoked inside the counted proof at source line 478 (a navigation-pointer stub, since the relation is a separate item of the same article). Supplies the sum formula which, after applying x^+_{theta-alpha}, yields part (2).

*Source reference.* Original reference garland: sources/sigma-2014-110-demazure-fusion/pinned-original-source.tex, source lines 315–315. Navigation pointer only: the counted text refers to the item garland of this same article, which is not part of this card. The pointer carries the item's own author label as its number; its content is not reproduced here.

**Lemma 1.** *Let $k \geq 1$ and $0 \leq i \leq k$. The following relations hold in the module $D(1,(k+1)\theta)$:*

1) $(x_\alpha^- \otimes 1)^{kd_\alpha+1}\big(x_\theta^- \otimes t^{2k+1-i}\big)\overline{w}_{(k+1)\theta} = 0$, $\forall\, \alpha \in R^+$, $(\theta|\alpha) = 1$;

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2) $\big(x_\alpha^- \otimes t^k\big)(x_\theta^- \otimes t^{2k+1-i})\overline{w}_{(k+1)\theta} \in \big\langle \big(x_\theta^- \otimes t^{2k+2-i}\big)\overline{w}_{(k+1)\theta}\big\rangle$, $\forall\, \alpha \in R^+$, $(\theta|\alpha) = 1$;
3) $\big(x_\theta^- \otimes t^{2k-i}\big)\big(x_\theta^- \otimes t^{2k+1-i}\big)\overline{w}_{(k+1)\theta} \in \big\langle \big(x_\theta^- \otimes t^{2k+2-i}\big)\overline{w}_{(k+1)\theta}\big\rangle$.

**Lemma 2.** *Let $k \in \mathbb{N}$. The following relations hold in the local Weyl module $W((k+1)\theta)$:*

1) $(x_\theta^- \otimes 1)^{2k+1}\big(x_\theta^- \otimes t^{2k+1-i}\big)w_{(k+1)\theta} = 0, \qquad \forall\, 0 \le i \le k;$
2) $(x_\theta^- \otimes t^m)(x_\theta^- \otimes t^{m+1})w_{(k+1)\theta} \in \big\langle \big(x_\theta^- \otimes t^{m+2}\big)w_{(k+1)\theta}\big\rangle, \qquad \forall\, m \ge k.$

*Proof.* Let $\alpha \in R^+$with$(\theta|\alpha) = 1$. This implies that $\theta - \alpha$ is also a root of $\mathfrak{g}$ and $(\theta|\theta - \alpha) = 1$. We now prove part (1). Observe that, $(x_\theta^- \otimes t^{2k+1-i})\overline{w}_{(k+1)\theta}$ is an element of weight $k\theta$. Further $(x_\alpha^+ \otimes 1)(x_\theta^- \otimes t^{2k+1-i})\overline{w}_{(k+1)\theta} = 0$, since $(x_\alpha^+ \otimes 1)\overline{w}_{(k+1)\theta} = 0$ and $(x_{\theta-\alpha}^- \otimes t^{2k+1-i})\overline{w}_{(k+1)\theta} = 0$, for all $0 \le i \le k$. Considering the copy of $\mathfrak{sl}_2$ spanned by $x_\alpha^+ \otimes 1$, $x_\alpha^- \otimes 1$, $\alpha^\vee \otimes 1$, we obtain part (1) by standard $\mathfrak{sl}_2$ arguments. We now prove part (2). Putting $r = 2$, $s = (3k + 1 - i)$ and $\alpha = \theta$ in Lemma garland, we get

$$\big(x_\theta^- \otimes t^k\big)\big(x_\theta^- \otimes t^{2k+1-i}\big)\overline{w}_{(k+1)\theta} + \sum_{\substack{k+1 \le p \le q \le 2k-i \\ p+q=3k+1-i}} \frac{1}{(2\delta_{p,q})!}\big(x_\theta^- \otimes t^p\big)\big(x_\theta^- \otimes t^q\big)\overline{w}_{(k+1)\theta}$$
$$\in \big\langle \big(x_\theta^- \otimes t^{2k+2-i}\big)\overline{w}_{(k+1)\theta}\big\rangle, \tag{1}$$

since $(x_\theta^- \otimes 1)^{3k+3-i}\overline{w}_{(k+1)\theta} = 0$, $\forall\, 0 \le i \le k$. Now we act on both sides of (1) by $x_{\theta-\alpha}^+$ and use the relation $(x_\alpha^- \otimes t^r)\overline{w}_{(k+1)\theta} = 0$, for all $r \ge (k+1)$, which gives part (2). Part (3) is immediate from the part (2) of Lemma 2. □